University of Tabuk
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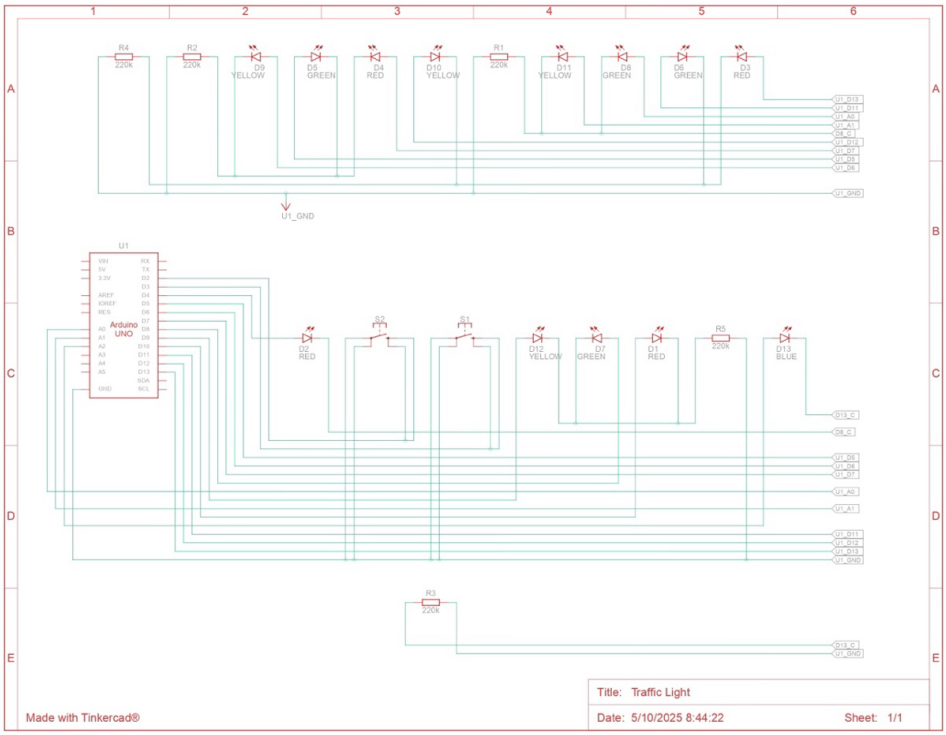
MICROPROCESSORS & EMBEDDED SYSTEMS LAB
CEN-412

Project (1). Traffic Light — 10 points

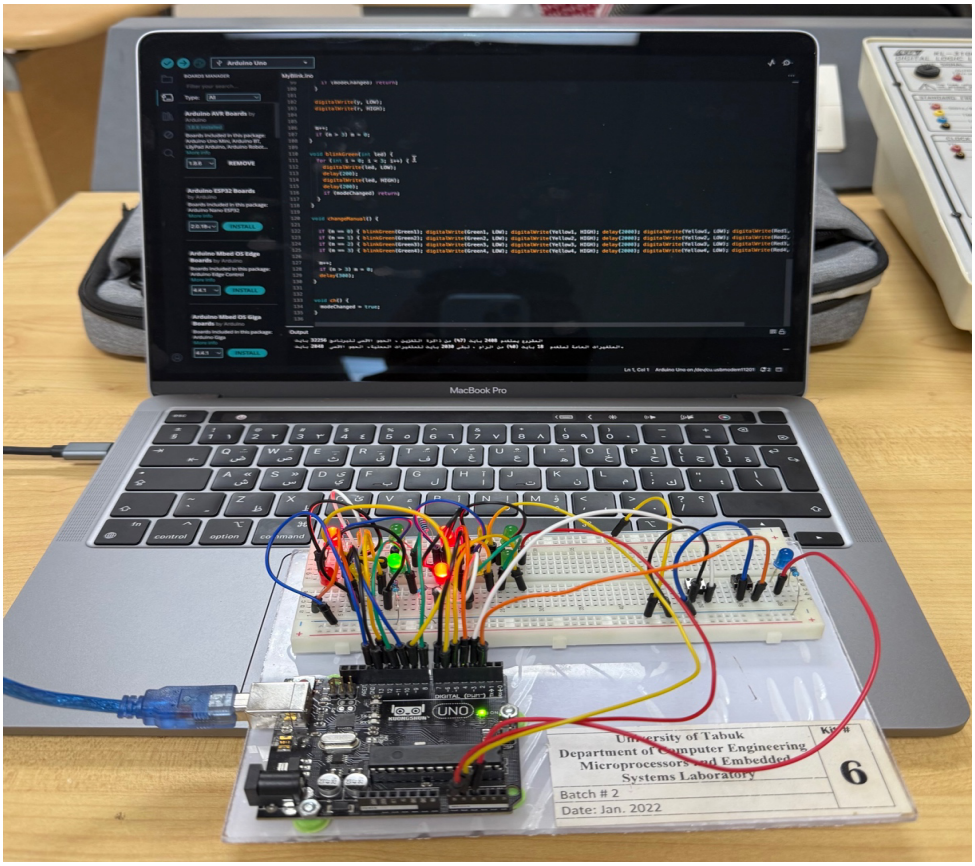
1. Project description

This project involves creating a system to control traffic lights at a four- way intersection with two modes of operation: automatic mode, where the traffic lights are controlled without human intervention, and manual mode, where a user can control the traffic lights using a switch. The manual mode is designed to provide emergency control of the traffic lights, allowing separate control of each street. The traffic lights are controlled using the intersection() function which determines the appropriate lighting for each traffic signal. Additionally, the ISR1 () function is used to toggle between manual and automatic modes using a switch, and ISR2 () function is used to change between different traffic light in different intersections.

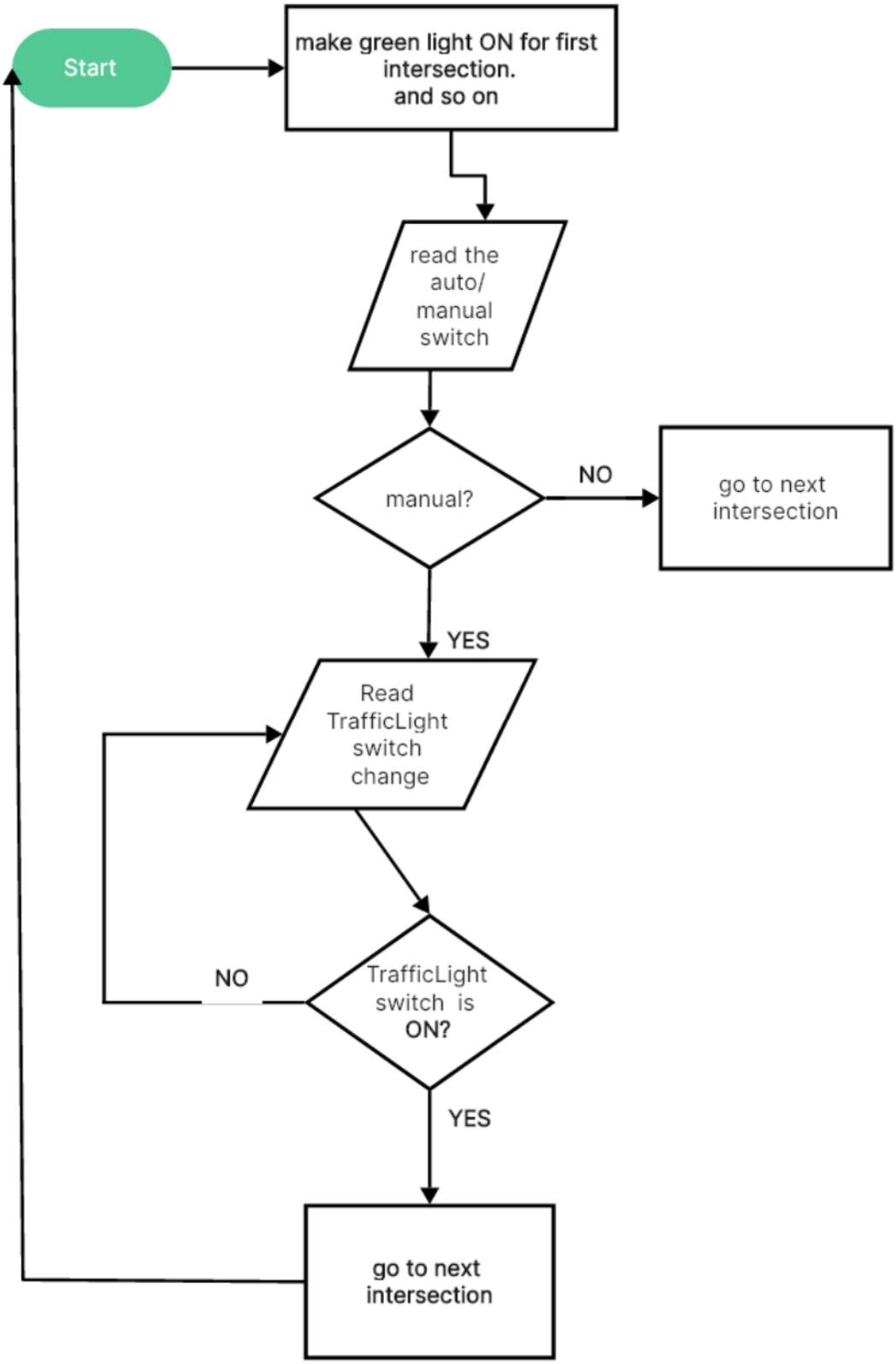
2. Project schematic Diagram



In Lab :



3. Algorithm and Program source code



```
int Green1 = 5, Yellow1 = 6, Red1 = 7;
int Green2 = 11, Yellow2 = 12, Red2 = 13;
int Green3 = 8, Yellow3 = 9, Red3 = 10;
int Green4 = A0, Yellow4 = A1, Red4 = 4;

int ModeButton = 2;
int ManualButton = 3;
int BlueLed = A2;

volatile bool mode = true;
volatile bool modeChanged = false;
int m = 0;

void setup() {
  pinMode(Green1, OUTPUT); pinMode(Yellow1, OUTPUT); pinMode(Red1, OUTPUT);
  pinMode(Green2, OUTPUT); pinMode(Yellow2, OUTPUT); pinMode(Red2, OUTPUT);
  pinMode(Green3, OUTPUT); pinMode(Yellow3, OUTPUT); pinMode(Red3, OUTPUT);
  pinMode(Green4, OUTPUT); pinMode(Yellow4, OUTPUT); pinMode(Red4, OUTPUT);

  pinMode(ModeButton, INPUT_PULLUP);
  pinMode(ManualButton, INPUT_PULLUP);
  pinMode(BlueLed, OUTPUT);

  attachInterrupt(digitalPinToInterrupt(ModeButton), ch, FALLING);

  allOff();

  digitalWrite(Red1, HIGH);
  digitalWrite(Red2, HIGH);
  digitalWrite(Red3, HIGH);
  digitalWrite(Red4, HIGH);
}

void loop() {
  if (modeChanged) {

    if (m == 0) autoSignal(Green1, Yellow1, Red1);
    else if (m == 1) autoSignal(Green2, Yellow2, Red2);
    else if (m == 2) autoSignal(Green3, Yellow3, Red3);
    else if (m == 3) autoSignal(Green4, Yellow4, Red4);
```

```
  } else {

    if (digitalRead(ManualButton) == LOW) {
      changeManual();
    }


    allOff();
    if (m == 0) { digitalWrite(Green1, HIGH); digitalWrite(Red2, HIGH); digitalWrite(Red3, HIGH);
digitalWrite(Red4, HIGH); }
    if (m == 1) { digitalWrite(Green2, HIGH); digitalWrite(Red1, HIGH); digitalWrite(Red3, HIGH);
digitalWrite(Red4, HIGH); }
    if (m == 2) { digitalWrite(Green3, HIGH); digitalWrite(Red1, HIGH); digitalWrite(Red2, HIGH);
digitalWrite(Red4, HIGH); }
    if (m == 3) { digitalWrite(Green4, HIGH); digitalWrite(Red1, HIGH); digitalWrite(Red2, HIGH);
digitalWrite(Red3, HIGH); }
  }
}

void allOff() {
  digitalWrite(Green1, LOW); digitalWrite(Yellow1, LOW); digitalWrite(Red1, LOW);
  digitalWrite(Green2, LOW); digitalWrite(Yellow2, LOW); digitalWrite(Red2, LOW);
  digitalWrite(Green3, LOW); digitalWrite(Yellow3, LOW); digitalWrite(Red3, LOW);
  digitalWrite(Green4, LOW); digitalWrite(Yellow4, LOW); digitalWrite(Red4, LOW);
}

void autoSignal(int g, int y, int r) {
  allOff();


  if (r != Red1) digitalWrite(Red1, HIGH);
  if (r != Red2) digitalWrite(Red2, HIGH);
  if (r != Red3) digitalWrite(Red3, HIGH);
  if (r != Red4) digitalWrite(Red4, HIGH);


  digitalWrite(r, LOW);
  digitalWrite(g, HIGH);

  for (int i = 0; i < 50; i++) {
    delay(100);
    if (modeChanged) return;
```

```
  }


  blinkGreen(g);
  digitalWrite(g, LOW);
  digitalWrite(y, HIGH);

  for (int i = 0; i < 20; i++) {
    delay(100);
    if (modeChanged) return;
  }

  digitalWrite(y, LOW);
  digitalWrite(r, HIGH);



  m++;
  if (m > 3) m = 0;
}

void blinkGreen(int led) {
  for (int i = 0; i < 3; i++) {
    digitalWrite(led, LOW);
    delay(200);
    digitalWrite(led, HIGH);
    delay(200);
    if (modeChanged) return;
  }
}

void changeManual() {


  if (m == 0) { blinkGreen(Green1); digitalWrite(Green1, LOW); digitalWrite(Yellow1, HIGH);
delay(2000); digitalWrite(Yellow1, LOW); digitalWrite(Red1, HIGH); }
  if (m == 1) { blinkGreen(Green2); digitalWrite(Green2, LOW); digitalWrite(Yellow2, HIGH);
delay(2000); digitalWrite(Yellow2, LOW); digitalWrite(Red2, HIGH); }
  if (m == 2) { blinkGreen(Green3); digitalWrite(Green3, LOW); digitalWrite(Yellow3, HIGH);
delay(2000); digitalWrite(Yellow3, LOW); digitalWrite(Red3, HIGH); }
  if (m == 3) { blinkGreen(Green4); digitalWrite(Green4, LOW); digitalWrite(Yellow4, HIGH);
delay(2000); digitalWrite(Yellow4, LOW); digitalWrite(Red4, HIGH); }


  m++;
```

```
  if (m > 3) m = 0;
  delay(300);
}


void ch() {
  modeChanged = !modeChanged;
  digitalWrite(BlueLed, modeChanged);
  delay(300);
}
```

4. Program description

- **The program begins by defining** the pin numbers for all LEDs and buttons:
    - Four intersections, each with Green, Yellow, and Red LEDs.
    - Two control buttons: one for mode switching (`ModeButton`) and one for manual operation (`ManualButton`).
    - One Blue LED to indicate the current mode (ON = Manual mode).
- **In the `setup()` function:**
    - All LED pins are set as **outputs**, and button pins are set as **inputs with pull-up resistors**.
    - An **interrupt** is attached to the `ModeButton` pin to detect mode changes immediately.
    - The function `allOff()` turns off all LEDs, and then all red lights are turned ON to start safely (all intersections are stopped).
- **The interrupt service routine `ch()`** is defined to handle mode changes:
    - When the Mode button is pressed, it sets `modeChanged = true`, which later toggles between **Automatic** and **Manual** operation.
- **Inside the `loop()` function:**
    - If `modeChanged` is true, the program switches between modes and adds a small delay for button debounce.
    - The Blue LED shows the current mode:
        - **OFF** → Automatic mode
        - **ON** → Manual mode
- **In Automatic Mode (`mode == true`):**
    - The program automatically runs traffic signals in order (from intersection 1 to 4).
    - Each intersection is controlled by the function `autoSignal()`.
    - Inside `autoSignal()`:
        - All lights are turned off.
        - The **green light** turns ON for 5 seconds.
        - It then **blinks 3 times** to warn that it's changing.
        - The **yellow light** turns ON for 2 seconds.
        - Finally, the **red light** turns ON.
        - The system moves to the next intersection (`m++`).
- **In Manual Mode (`mode == false`):**
    - When the **Manual Button** is pressed:
        - The current intersection's **green light blinks** 3 times.
        - The **yellow light** turns ON for 2 seconds.
        - The **red light** turns ON afterward.
        - The system moves to the next intersection (`m++`).

- Only one intersection's **green light** stays ON at a time, depending on the current value of `m`.

- **Helper functions:**

  - `allOff()`: turns off all LEDs across all intersections.
  - `blinkGreen(led)`: blinks the given green LED 3 times (each blink 300 ms).
  - `autoSignal(g, y, r)`: handles one automatic light sequence (green → blink → yellow → red).